May 19 (Thu), 2022
Advanced Calculus

Mid-term Exam

NOTE: Show explicitly how you get the answers. Do not skip lines!

Prob 1(10) Show that the polar equation $r = a\sin\theta$ represents a circle where $a \neq 0$, and find its center and radius.

Prob 2(20) Sketch the curve for $r = \dfrac{1}{\theta}$ for $\dfrac{\pi}{2} \leq \theta \leq 2\pi$. Also, find the length of the curve.
(Hint) You may use the follwoing formula.

$$\int \frac{1}{\sqrt{t^2+1}} dt = \sinh^{-1} t + C$$

Prob 3(10) Sketch the curve for $r = \sqrt{\sin(2\theta)}$ for $0 \leq \theta \leq \dfrac{\pi}{2}$ and find the area enclosed by the curve.

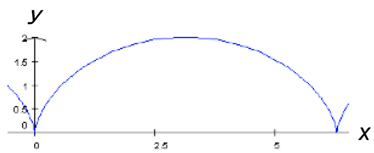
Prob 4(10) Let us consider a curve whose vector equation is given by

$$\mathbf{r}(t) = e^{-t}\cos t\,\mathbf{i} + e^{-t}\sin t\,\mathbf{j} + a\,\mathbf{k}$$

where a is constant. Using the fact that an infinitesimal length of the curve is expressed by $ds = \sqrt{d\mathbf{r}\cdot d\mathbf{r}}$, find the length of the curve for $0 \leq t \leq 2\pi$.

Prob 5(20) Find the smallest radius of curvature of the function $y = \exp(x)$.

Prob.6(30) A curve of cycloid (right figure) is expressed by the follwoing parametric equations:



$$x = a(t - \sin t), \quad y = a(1 - \cos t)$$

where t $(0 \leq t \leq 2\pi)$ is a parameter.

(a) Show that the unit tangent vector $\mathbf{T}(t)$ to the cycloid is expressed by

$$\mathbf{T}(t) = \sin\left(\frac{t}{2}\right)\mathbf{i} + \cos\left(\frac{t}{2}\right)\mathbf{j}$$

(b) Find the normal vector $\mathbf{N}(t)$ and binormal vector $\mathbf{B}(t)$ on the cycloid.

(c) Find the arc-length $s(t)$ of the cycloid from the origin as a function of t.